

Enhancement of a User Authentication Scheme for Big Data Collection in IoT-Based Intelligent Transportation System

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Abstract—In recent years, research on the Internet of Things (IoT) has seen widespread growth, with a particular focus on mobile networks, especially intelligent transportation systems (ITS), due to their potential application in smart cities. However, communication within ITS suffers from numerous issues, leading to multiple attacks. Therefore, ensuring the secure transmission of messages in ITS poses a significant challenge. Recently, Srinivas et al. have proposed "a three-factor authentication scheme for Big Data collection in IoT-based ITS, called UAP-BCIoT". However, the proposed scheme has some shortcomings in terms of authentication. In this paper, we show how this flaw affects the users' access to the data provided by the IoT devices, which are deployed on vehicles. In addition, we propose enhancements to overcome the scheme's authentication issue. Furthermore, a comparative analysis reveals that the enhanced UAP-BCIoT achieves better efficiency, security, and performance compared to UAP-BCIoT and other schemes.

Index Terms—Secure Big Data collection, authentication, key agreement, cryptanalysis, IoT-based ITS.

I. INTRODUCTION

BY 2050, over 70% of the global population is projected to reside in urban areas. The rapid pace of urbanization will give rise to a diverse range of organizational and technical challenges [2], [3]. In order to effectively manage urbanization growth, numerous countries have embraced the concept of smart cities including Intelligent Transportation System (ITS). Generally, the term "ITS" refers to the utilization of technology-based methods, such as the Internet of Things (IoT), cyber-physical systems [5], and big-data analysis, with the aim of improving and enhancing the quality of life for citizens [4]. There are various network communication systems associated with ITS [6], [8], such as:

- *Intra-Vehicle communication*: This communication system monitors the internal performance of a smart vehicle by leveraging its components, including installed IoT devices.
- *Vehicle-to-Vehicle communication (V2V)*: The V2V communication enables the exchange of information between

smart vehicles, such as the position and speed of nearby vehicles [7].

- *Vehicle-to-Infrastructure communication (V2I)*: The V2I communication enables the exchange of information between smart vehicles and ITS infrastructure (Road side unit).
- *Vehicle-to-smart grid communication (V2G)*: The exchange of information between smart vehicles and the smart grid (SG) facilitates features such as grid balancing, vehicle charging, and energy trading.
- *Vehicle-to-Cloud Communication (V2C)*: The information and data exchange between vehicles and the cloud enables different services and applications such as remote vehicle monitoring, location information, and vehicle diagnostics.
- *Vehicle-to-User Communication (V2U)*: V2U communications refers to the transmission of data between vehicles and individuals, facilitating remote vehicle control, tracking location, and real-time vehicle diagnostics for ensuring efficient vehicle maintenance and management.

The safety protection of critical end-point devices is a significant concern, as any breach by hackers could result in potential hazards and malfunctions in vehicles. To mitigate these risks, it is crucial to implement a secure communication design that effectively restricts hackers from creating dangerous malfunctions. Stated differently, when designing ITS system, we must take into account resilience against attacks such as privileged-insider, impersonation, replay and man-in-the-middle. In general, authentication is widely regarded as a reliable solution for securing communications. Additionally, it is imperative to avoid exposing the vehicle's identity. Therefore, an authentication scheme designed for ITS must incorporate three other essential features: untraceability, mutual authentication, and anonymity. Numerous authentication schemes have been proposed in academic literature.

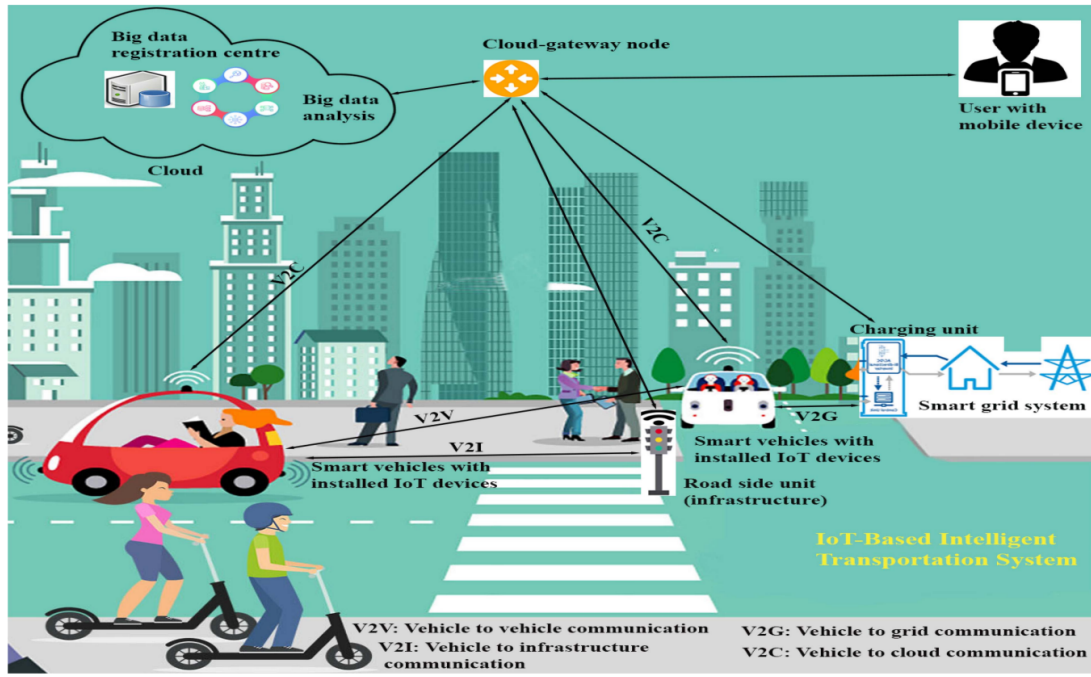


Fig. 1: Network model of UAP-BCIoT scheme

The remainder of this paper is organized as follows. The discussion of the related work is presented in Section II. Section III reviews and analysis UAP-BCIoT scheme. We will discuss why this flaw occurred and how it affects the paper assertions in Section IV. we provide Enhanced version of UAP-BCIoT in Section V. A detailed communication and computation cost analysis for the Enhanced UAP-BCIoT is presented in Section VI. Finally, we summarize this work in Section VII.

II. RELATED WORK

Turkanoviá et al. [9] proposed two-factor user authentication protocol that is applicable for the IoT environment, Unfortunately, their protocol is susceptible to a number of well-known attacks, such as privileged-insider attack, impersonation attack, stolen smart-card attack and off-line password guessing attack.

Porambage et al. [10] developed a user authentication mechanism for IoT deployments that allows end user to authenticate directly to IoT sensing nodes, granting them access to sensing information and services. However, despite its computational efficiency, this scheme is exposed to multiple known vulnerabilities, such as main-in-the-middle attack and IoT node capture attack. Li et al. [11] introduced an authentication scheme based on elliptic curve cryptography (ECC) for an industrial IoT environment. However, a significant limitation of their scheme is its inability to guarantee certain functionality attributes. Additionally, the high computational cost associated with the execution of this authentication scheme makes it impractical for IoT sensing nodes with limited resources.

In 2019, Banerjee et al. [13] presented a "three-factor user authentication scheme" for the deployment of IoT devices in a general context. Their scheme utilized symmetric key encryption techniques to establish a session key between users and IoT smart devices. However, their proposal lacks support for the crucial "big data analytics phase," limiting its suitability for efficient data collection and analysis in ITS that is based on IoT. Additionally, it does not address the validation of IoT device credentials by the gateway once the IoT devices are deployed in the network.

Recently, Srinivas et al. [1] proposed "Secure User Authentication Protocol for Big Data Collection in IoT-Based Intelligent Transportation System called UAP-BCIoT" addresses the problem of security protection for IoT smart devices in the context of Intelligent Transportation Systems (ITS). It is this protocol that we have chosen to study in this paper. In this scheme for BCIoT based ITS environment, the Trusted Big Data Registration Center (*BRC*) is responsible for registering the semi-trusted Cloud-Gateway (*CG*) node, users, and smart devices. The Cloud-Gateway device in UAP-BCIoT is used for mutual authentication between the user and an IoT node. Following successful authentication, a legitimate user and an IoT node can exchange a secret session key for real-time device information access, ensuring data security and integrity. UAP-BCIoT can assure protection from a number of well-known attacks. In addition, their scheme is perfect for collecting and analyzing Big Data in IoT-based intelligent transportation systems, but it contains a flaw. This flaw affects the authentication between honest users and IoT devices deployed inside vehicles.

This paper makes the following contributions:

- Initially, the review UAP-BCIoT scheme are presented.
- It is proved that the UAP-BCIoT contains a flaw that prevents users from accessing IoT devices.
- An enhanced scheme is proposed to ensure user access to IoT devices, considering efficiency and security.
- Finally, enhanced UAP-BCIoT is compared with other relevant schemes, taking into account performance and security features as key considerations.

III. REVIEWS OF UAP-BCIoT SCHEME

In this section, we review the recently presented UAP-BCIoT by Srinivas et al. [1]. To facilitate the description and analysis of the UAP-BCIoT scheme, we review the network model they use. Figure 1 depicts a typical ITS entities, where users access IoT devices via the public internet. These entities primarily encompass users, the cloud-gateway node, and the IoT nodes, but it also contains BRC, smart vehicles, vehicle charging units and roadside units. In this network model, various types of communications are present: vehicle-to-vehicle, vehicle-to-infrastructure, vehicle-to-grid and vehicle-to-cloud communication. UAP-BCIoT focuses on the communication between authorized users who can access data from IoT nodes directly installed in intelligent vehicles via the CG node. Their scheme comprises of different phases. The commonly used notations throughout this paper are listed in Table 1.

A. System Initialization Phase

The BRC serves as the authorized entity within the network responsible for retrieving diverse parameters for each deployed node. Additionally, the authorized Cloud-Gateway (CG) selects its own private and public keys along with other parameters.

B. IoT device enrollment phase

The BRC conducts this phase offline. The BRC computes the secret $IC_{j_1} \leftarrow h(SK_{CG-BRC} \parallel ID_{IN_j})$ for each IoT node (ID_{IN_j}). The BRC saves (ID_{IN_j}, IC_{j_1}) in the memory of IN_j prior to its deployment within the ITS environment.

C. User Registration Phase

In order to register and to become a legitimate user, U_i executes the following steps:

- Initially, the user U_i has the freedom of picking his/her identity (ID_i) and password (PW_i) and determines its bio-metric identity information (BIO_i). Thereafter, U_i calculates $Gen(BIO_i) \leftarrow (\sigma_i, \tau_i)$, $MID_i \leftarrow h(ID_i \parallel b_i)$ and $MPW_i \leftarrow h(PW_i \parallel \sigma_i)$ where b_i is random number generated by MD_i . After that, U_i forward $\{MID_i, MPW_i\}$ along with a registration request to CG_k via a secure channel.
- Upon reception of request message, CG_k computes $G_1 \leftarrow (g_{pri} \cdot h(MID_i)) \cdot P$, $G_2 \leftarrow G_1 \oplus h(MPW_i \parallel MID_i)$ and $G_3 \leftarrow G_1 \oplus h(X_{CG_k})$. Thereafter, CG_k forward $\{G_1, G_2, G_3, h(\cdot), P\}$ to U_i via secure channel.
- On the reception of reply message, the U_i calculates $L_i \leftarrow b_i \oplus h(ID_i \parallel \sigma_i \parallel PW_i)$, $G_2^* \leftarrow G_2 \oplus h(b_i \parallel \sigma_i \parallel$

$$PW_i), G_3^* \leftarrow G_3 \oplus h(\sigma_i \parallel b_i \parallel PW_i), G_4 \leftarrow h(G_1 \parallel PW_i \parallel b_i \parallel \sigma_i).$$

- Finally, U_i keeps $\{L_i, G_2^*, G_3^*, G_4, h(\cdot), Gen(\cdot), Rep(\cdot), \tau_i, P\}$ in its mobile device.

TABLE I: COMMONLY USED NOTATIONS

Notations	Elucidations
BRC	Trusted Big data Registration Center
CG_k	k^{th} semi-trusted Cloud-Gateway
IN_j	j^{th} IoT smart device
U_i, PW_i	i^{th} user and its password
ID_{CG_k}, ID_{IN_j}	The identification of CG_k and IN_j , respectively
X_{CG_k}	Secret key of CG_k
P	ECC point generator.
$G_{pub}=G_{pri} \cdot P$	Public key computed by CG_k
x, y	Arbitrary numbers generated by U_i and IN_j , respectively
TS_1, TS_2, TS_3	Time-stamps

D. Login and Authentication Phase

Suppose that at some point, a user intends to monitor the data from a specific IoT node integrated in a intelligent vehicle. To ensure user privacy and maintain the authenticity of the acquired data, it is crucial for both users and IoT nodes to confirm the identities of each other and establishment of a shared secret key prior to exchanging any data. Fig. 2 illustrates the comprehensive process of login and authentication phase of UAP-BCIoT.

IV. PROBLEM STATEMENT

The login and authentication phase of UAP-BCIoT is divided into five steps. The authentication process is carried out by the cloud-gateway device (CG_k) to validate the participating users and determine whether an IoT node (IN_j) is operating correctly or not in the network. The problem appears in steps LA2 and LA3.

The step LA2 of login and authentication phase of the considered manuscript is as follows. The mobile device (MD_i) verifies G_4 . If correct, MD_i confirms that the user (U_i) entered credentials are valid.

U_i selects an accessed IoT device identity, then generates a random nonce (x) and a timestamp (TS_1) to calculate the following values:

$$M_x \leftarrow x \cdot P \quad (1)$$

$$HID_i \leftarrow MID_i^* \oplus h((G_1 \oplus G_3) \parallel TS_1) \quad (2)$$

$$A_1 \leftarrow x \cdot (G_{pub} + h(ID_{IN_j}) \cdot P) \quad (3)$$

$$A_2 \leftarrow A_1 \oplus h(G_2 \parallel G_1 \parallel M_x \parallel TS_1) \quad (4)$$

$$A_3 \leftarrow h((G_1 \oplus G_3) \parallel MID_i^* \parallel TS_1 \parallel A_2) \quad (5)$$

$$DID_{IN_j} \leftarrow ID_{IN_j} \oplus h(G_1 \parallel TS_1) \quad (6)$$

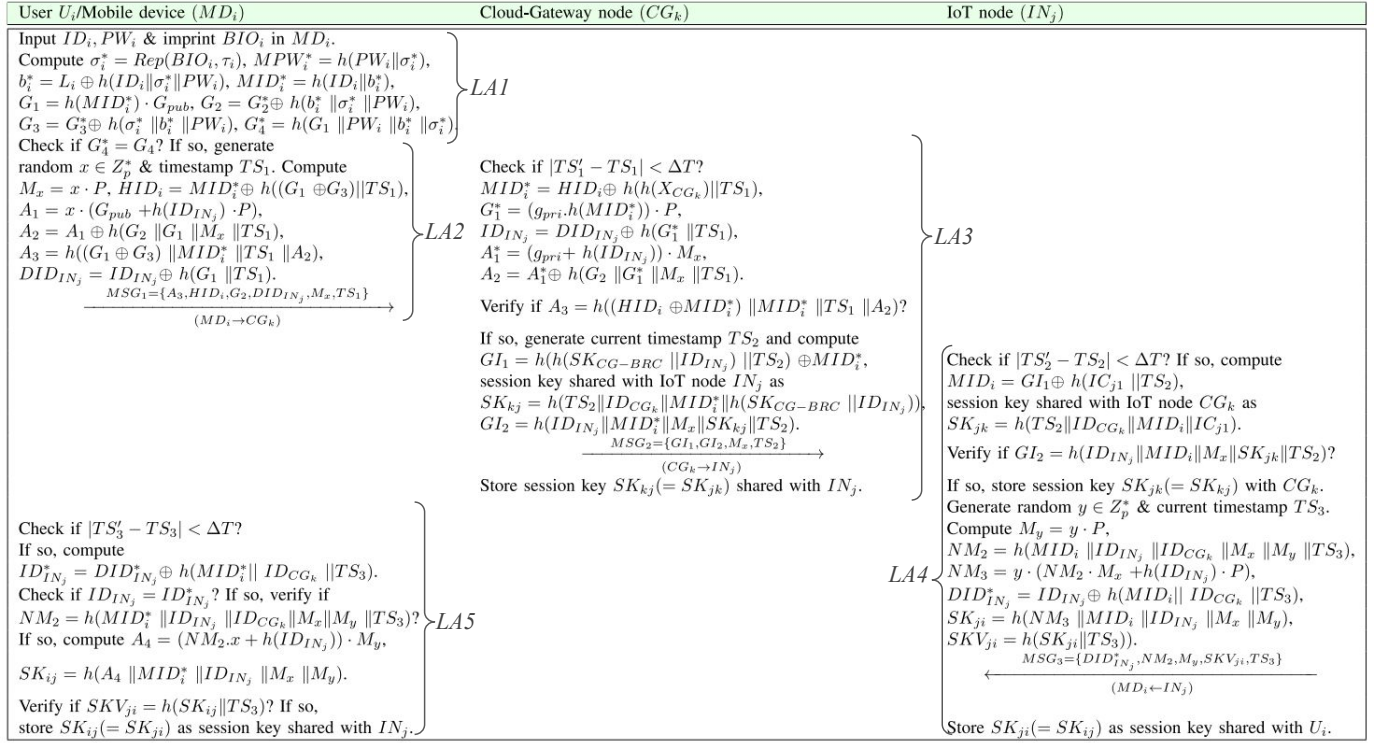


Fig. 2: Authentication phase of UAP-BCIoT-scheme

The mobile device sends the request message (MSG_1) to the semi-trusted Cloud-Gateway (CG_k) over a public channel.

$$MSG_1 \leftarrow \{A_3, HID_i, G_2, DID_{IN_j}, M_x, TS_1\}.$$

In step LA3, CG_k verifies the maximum transmission delay after receiving the request message MSG_1 . If the verification is successful, CG_k computes

$$MID_i^* \leftarrow HID_i \oplus h(h(X_{CG_k}) \parallel TS_1) \quad (7)$$

$$G_1^* \leftarrow (g_{pri} \cdot h(MID_i^*)) \cdot P \quad (8)$$

$$ID_{IN_j} \leftarrow DID_{IN_j} \oplus h(G_1^* \parallel TS_1) \quad (9)$$

$$A_1^* \leftarrow (g_{pri} + h(ID_{IN_j})) \cdot M_x \quad (10)$$

$$A_2 \leftarrow A_1^* \oplus h(G_2 \parallel G_1^* \parallel M_x \parallel TS_1) \quad (11)$$

After the above calculation, CG_k verifies:

$$A_3 \stackrel{?}{=} h((HID_i \oplus MID_i^*) \parallel MID_i^* \parallel TS_1 \parallel A_2) \quad (12)$$

This check always fails, then the session is called off even if the message received from the user is legitimate. The reason for this failure is the flaw in the description of the verification value (A_3) computed by U_i Eq. (5) and verified by CG_k Eq. (12) are not equivalent, which is justified by the following:

$$A_3 \text{ of the } U_i = h((G_1 \oplus G_3) \parallel MID_i^* \parallel TS_1 \parallel A_2)$$

$$\begin{aligned} &= h((G_1 \oplus (G_1 \oplus h(X_{CG_k}))) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((G_1 \oplus G_1 \oplus h(X_{CG_k})) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((\emptyset \oplus h(X_{CG_k})) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(X_{CG_k}) \parallel MID_i^* \parallel TS_1 \parallel A_2) \end{aligned}$$

$$\begin{aligned} A_3 \text{ of the } CG_k &= h((HID_i \oplus MID_i^*) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((MID_i^* \oplus h((G_1 \oplus G_3) \parallel TS_1)) \oplus MID_i^*) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(((h((G_1 \oplus G_1 \oplus h(X_{CG_k}) \oplus \emptyset) \parallel TS_1))) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(((h((\emptyset \oplus h(X_{CG_k})) \parallel TS_1))) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(h(X_{CG_k}) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &\neq A_3 \text{ of the } U_i \end{aligned}$$

The above-mentioned problem (CG_k -side and U_i -side) has rendered A_3 computed by U_i and verified by CG_k not equivalent. Without this equivalency, the paper's main claims, such as the authentication of a legitimate user by Cloud-Gateway node, are not valid.

V. THE PROPOSED VERSION

In this section, we propose an enhanced version of UAP-BCIoT scheme to overcome these flaws. Two possible remedies are proposed in the following sub-sections.

A. Cloud-Gateway side remedy

The cloud-gateway device (CG_k) conducts the authentication process to validate the participating users, CG_k verifies

the validity of the participant U_i by checking A_3 after receiving the login request MSG_1 .

We have shown that A_3 computed by U_i and verified by CG_k are not equivalent. This led a legitimate user to be suspended. Nevertheless, our discussion points out that there is a way to guarantee equivalence between A_3 received from a user of the form Eq. (5) and A_3 verified by CG_k . For that reason, we adjust Step LA2 Eq. (2) and Step LA3 Eq. (7) as follows:

$$Eq.(2) : HID_i \leftarrow MID_i^* \oplus (G_1 \oplus G_3) \quad (13)$$

$$Eq.(7) : MID_i^* \leftarrow HID_i \oplus h(X_{CGK}) \quad (14)$$

In this way, the verification value A_3 computed by U_i and A_3 computed by CG_k are equivalent, which is justified by the following:

$$\begin{aligned} A_3 \text{ of the } CG_k &= h((HID_i \oplus MID_i^*) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((MID_i^* \oplus (G_1 \oplus G_3)) \oplus MID_i^* \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((MID_i^* \oplus (G_1 \oplus G_3) \oplus MID_i^*) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((G_1 \oplus G_1 \oplus h(X_{CGK})) \oplus \phi \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h((\phi \oplus h(X_{CGK})) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(X_{CGK}) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= A_3 \text{ of the } U_i \text{ (UAP-BCIoT)} \end{aligned}$$

With the first solution (CG_k -side), the issue of Authentication of legitimate users to the desired IoT devices is solved.

B. User side remedy

Authorized external users can access the data from IoT sensors directly deployed in vehicles through the login request message (MSG_1) to the cloud-gateway device. In the previous section, we indicated the equality problem of Eq. (5) and Eq. (12). In this subsection, we provide the correct A_3 (Eq. 5) for the login request message. More detail in the following:

$$Eq.(5) : A_3 \leftarrow h(h((G_1 \oplus G_3) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \quad (15)$$

In this way, the verification A_3 computed by U_i and A_3 computed by CG_k are equivalent, which is justified by the following:

$$\begin{aligned} A_3 \text{ of the } U_i &= h(h(G_1 \oplus G_3 \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(G_1 \oplus (G_1 \oplus h(X_{CGK}))) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(G_1 \oplus G_1 \oplus h(X_{CGK})) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(\phi \oplus h(X_{CGK})) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= h(h(h(X_{CGK}) \parallel TS_1) \parallel MID_i^* \parallel TS_1 \parallel A_2) \\ &= A_3 \text{ of the } CG_k \text{ (UAP-BCIoT)} \end{aligned}$$

With this second solution (U_i -side), the issue of authentication to the desired IoT devices is again solved.

VI. PERFORMANCE ANALYSIS

In this section, we perform a detailed comparison of UAP-BCIoT [1] with CG_k -side remedy, and U_i -side remedy re-

TABLE II: Security and Functionality Comparison

Protocol	Li [11]	Por [10]	Ba [13]	UAP-BCIoT [1]	Enhanced A UAP-BCIoT	Enhanced B UAP-BCIoT
A1	✓	×	✓	✓	✓	✓
A2	✓	×	×	✓	✓	✓
A3	✓	—	✓	✓	✓	✓
A4	✓	×	✓	✓	✓	✓
A5	✓	×	×	✓	×	✓
A6	✓	×	✓	✓	✓	✓
A7	✓	×	✓	✓	✓	✓
A8	✓	×	✓	✓	✓	✓
A9	×	×	×	✓	✓	✓
A10	×	×	×	✓	✓	✓
A11	✓	✓	✓	×	✓	✓
A12	✓	✓	✓	✓	✓	✓
A13	✓	✓	×	×	✓	✓

A1: "Privileged-insider attack"; A2: "User anonymity"; A3: "Stolen mobile /smart card"; A4: "Main-in-the-middle attack"; 5: "Replay attack"; A6: "Impersonation attacks"; A7: "DoS attack"; A8: "IoT node capture attack"; A9: "IoT node credential validation"; A10: "Big Data analytics"; A11: "Mutual authentication"; A12: "Traceability"; A13: "Accessibility × does not resist an attack or not supports an attribute" ✓ "resists an attack or supports an attribute"; —: "Not applicable in a scheme"

garding the security features, communication costs, and computation costs is shown. To do so, we based our analysis on the experimental findings of [1].

A. Comparison of Security

Srinivas et al. proved the security of their designed protocol (UAP-BCIoT) against various security attacks and its superior security measures and extensive functionality features compared to related schemes (Li et al. [11], Porambage et al. [10] and Banerjee et al. [13]). Our improved version (A) solves the user authentication problem but contains a replay attack. However, our enhanced version (B), which adds hash functions to the UAP-BCIoT protocol, ensures an equivalent level of security and functionality and fixes the users' authentication flaw in the UAP-BCIoT. We believe the second solution (section IV-B) to be the most appropriate. Table II offers a comprehensive comparison of the security and functionality features of the enhanced UAP-BCIoT and three other schemes. It can be seen that the enhanced UAP-BCIoT offers improved functionality features compared to UAP-BCIoT [1] and enhanced security features compared to the other three schemes.

TABLE III: RUNNING TIME FOR VARIOUS OPERATIONS

Notations	Description	Execution Time
T_h	Hash Function	0.32 ms
T_{sed}	Symmetric en/decryption	8.7 ms
T_{ecm}	ECC point multiplication	17.1 ms
T_{eca}	ECC point addition	4.4 ms
T_{fe}	Fuzzy Extractors	17.1 ms

B. Comparison of Computation Costs

The experimental results listed in Table III are used to calculate the executing time of various cryptographic operations. The computation cost for UAP-BCIoT, Enhanced UAP-BCIoT

TABLE IV: Performance Comparison

Performance Parameters		Li et al. [11]	Banerjee et al. [13]	UAP-BCIoT [1]	Enhanced UAP-BCIoT (A)	Enhanced UAP-BCIoT (B)
Computation Costs (in ms)	User	70,96	47,04	125,14	124,82	125,46
	Gateway	19,34	45,1	42,12	41,8	42,12
	IoT node	18,38	18,04	57,94		
		$19T_h + 6T_{ecm}$	$19T_h + 10T_{sed}$	$35T_h + 11T_{ecm}$	$33T_h + 11T_{ecm}$	$36T_h + 11T_{ecm}$
	Total Cost	108.68	$+T_{fe}= 159.58$	$+T_{fe} + 2T_{eca}= 225.20$	$+T_{fe} + 2T_{eca}= 224.56$	$+T_{fe} + 2T_{eca}= 225.52$
Communication Costs (Bits)	MSG_1	800	192	1152	992	1312
	MSG_2	640	1088	672		
	MSG_3	640	1024	832		
	MSG_4	640	-			
	Total Cost	2720	2304	2656	2496	2816

A and B represent 225.20, 224.56 and 225.52 milliseconds, respectively. In the previous section we decided to take version B to avoid security attacks. The enhanced UAP-BCIoT (B) guarantees security and fixes the flaw in the UAP-BCIoT with slightly higher computation costs compared to UAP-BCIoT.

C. Comparison of Communication Costs

For the computation of communication cost, it is assumed that identity, timestamp, hash function output (SHA-1), certificate (signature using ECDSA), and a random nonce require 160, 32, 160, 320, and 160 bits, respectively [1]. In the enhanced UAP-BCIoT, three exchanged messages $MSG_1 \leftarrow \{A_1, HID_i, G_2, DID_{IN_j}, M_x, TS_1\}$, $MSG_2 \leftarrow \{GI_1, GI_2, M_x, TS_2\}$ and $MSG_3 \leftarrow \{DID_{IN_j}^*, NM_2, M_y, SKV_{ji}, TS_3\}$ require 1312 Bits, 672 Bits and 832 Bits respectively. Table IV presents a summary of the number of exchanged messages and the communication costs for all the schemes. It is clear from Table IV that the enhanced UAP-BCIoT (B) has slightly higher computation costs compared to the UAP-BCIoT, which fixes the flaw in the UAP-BCIoT.

VII. CONCLUDING REMARKS

In this paper, we propose a new authentication protocol based on the existing UAP-BCIoT [1]. First show that UAP-BCIoT scheme lacks the authentication property on the user side. An enhanced scheme is subsequently proposed to address the weaknesses found in UAP-BCIoT with slightly higher communication & computation costs compared to UAP-BCIoT, it offers improved functionality features compared to UAP-BCIoT and enhanced security features compared to the other three schemes. In the future, our aim is to conduct testing and ongoing refinements of our solution within a real ITS network environment.

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